Computer Programming (CSN11101)
ASSIGNMENT-4

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Q1. Write a Program to create a pyramid pattern using C.
A1. CODE:

```
#include <stdio.h>

int main() {
    int rows,i,j,space;
    printf("Enter number of rows: ");
    scanf("%d", &rows);
    for (i=1;i<=rows;i++) {
        for(space=1;space<=rows-i;space++)
            printf(" ");
        for (j=1;j<=(2*i-1);j++)
            printf("*");
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/Computer programming/Assignment-4$ ./p1
Enter number of rows: 5
    *
   ***
  *****
 *******
*********
```

Q2. Write a program to Print Half half-pyramid pattern.
A2. CODE:

```
#include <stdio.h>

int main() {
    int rows,i,j;
    printf("Enter number of rows: ");
    scanf("%d",&rows);
    for (i=1;i<=rows;i++) {
        for(j=1;j<=i;j++){
            printf("%d",i);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/Computer programming/Assignment-4$ ./p2
Enter number of rows: 5
1
22
333
4444
55555
```

Q3. Write a program to find the value of one number raised to the power of another number (do not use pow() function).

A3. CODE:

```
#include <stdio.h>

int main(){
    int base,exp,i;
    long long result=1;
    printf("Enter base: ");
    scanf("%d",&base);
    printf("Enter exponent: ");
    scanf("%d",&exp);
    for(i=1;i<=exp;i++){
        result*=base;
    }
    printf("%d^%d = %lld\n",base,exp,result);
    return 0;
}
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/Computer programming/Assignment-4$ ./p3
Enter base: 6
Enter exponent: 3
6^3 = 216
```

Q4.Write a program that repeatedly asks the user for input until a specific value is entered to break the loop.

A4. CODE :

```
#include <stdio.h>

int main() {
    int num;
    printf("Enter numbers (enter -1 to stop):\n");
    while(1){
        scanf("%d",&num);
        if(num==-1){
            printf("Loop terminated.\n");
            break;
        }
        printf("You entered: %d\n",num);
    }
    return 0;
}
```

OUTPUT :



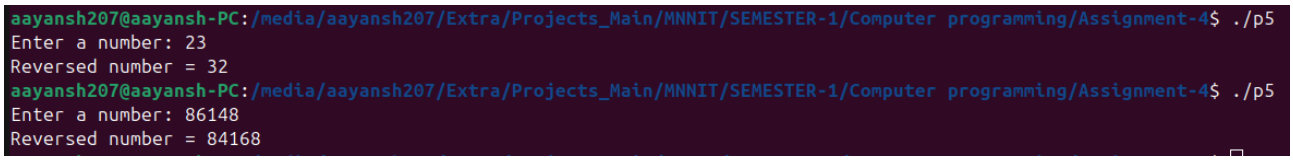
Q5. Write a program to obtain the reversed number of a given number. maximum digits in number are 7.

A5. CODE :

```
#include <stdio.h>

int main() {
    int num,rev=0,rem;
    printf("Enter a number: ");
    scanf("%d",&num);
    while (num!=0) {
        rem=num%10;
        rev=rev*10+rem;
        num/=10;
    }
    printf("Reversed number = %d\n",rev);
    return 0;
}
```

OUTPUT :



Q6.Write a program to print the first N numbers in the Fibonacci series.
A6. CODE:

```
#include <stdio.h>

int main() {
    int n,i;
    long long a=0,b=1,next;
    printf("Enter N (number of terms): ");
    scanf("%d",&n);
    printf("Fibonacci Series: ");
    for (i=1;i<=n;i++) {
        printf("%lld ",a);
        next=a+b;
        a=b;
        b=next;
    }
    printf("\n");
    return 0;
}
```

OUTPUT:

